



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP3623

Database Programming

2025/2026 SEMESTER 1

Lab Assignment 2

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Part A

Question A1

```
> db.createCollection("appointments")  
< { ok: 1 }  
> db.createCollection("loginSessions")  
< { ok: 1 }
```

Question A2

```
> db.appointments.insertMany([
  {
    appointmentId: "A001",
    patientName: "Zara",
    doctor: "Dr. Husin",
    department: "Pediatrics",
    status: "COMPLETED",
    fee: 200,
    appointmentDate: ISODate("2025-01-05T09:00:00Z")
  },
  {
    appointmentId: "A002",
    patientName: "Aisy",
    doctor: "Dr. Alif",
    department: "Cardiology",
    status: "PENDING",
    fee: 120,
    appointmentDate: ISODate("2025-01-06T10:00:00Z")
  },
  {
    appointmentId: "A003",
    patientName: "Nur",
    doctor: "Dr. Lee",
    department: "Dermatology",
    status: "COMPLETED",
    fee: 300,
    appointmentDate: ISODate("2025-01-07T11:00:00Z")
  },
],
```

```
{
  appointmentId: "A004",
  patientName: "Ain",
  doctor: "Dr. Qawie",
  department: "Neurology",
  status: "CANCELLED",
  fee: 150,
  appointmentDate: ISODate("2025-01-08T09:30:00Z")
},
{
  appointmentId: "A005",
  patientName: "Farhan",
  doctor: "Dr. Hana",
  department: "Neurology",
  status: "COMPLETED",
  fee: 150,
  appointmentDate: ISODate("2025-01-09T14:00:00Z")
},
{
  appointmentId: "A006",
  patientName: "Arisa",
  doctor: "Dr. Husin",
  department: "Cardiology",
  status: "COMPLETED",
  fee: 100,
  appointmentDate: ISODate("2025-01-10T15:00:00Z")
},
],
```

```
{
  appointmentId: "A007",
  patientName: "Fauzan",
  doctor: "Dr. Lee",
  department: "Dermatology",
  status: "PENDING",
  fee: 300,
  appointmentDate: ISODate("2025-01-11T08:30:00Z")
},
{
  appointmentId: "A008",
  patientName: "Nabil",
  doctor: "Dr. Alif",
  department: "Neurology",
  status: "COMPLETED",
  fee: 150,
  appointmentDate: ISODate("2025-01-12T10:30:00Z")
},
```

```
{
  appointmentId: "A009",
  patientName: "Shai",
  doctor: "Dr. Lee",
  department: "Neurology",
  status: "CANCELLED",
  fee: 160,
  appointmentDate: ISODate("2025-01-13T13:00:00Z")
},
{
  appointmentId: "A010",
  patientName: "Tasha",
  doctor: "Dr. Qawie",
  department: "Pediatrics",
  status: "COMPLETED",
  fee: 200,
  appointmentDate: ISODate("2025-01-14T16:00:00Z")
}
])
```

```
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('696d664a7431812aaf8cc852'),  
    '1': ObjectId('696d664a7431812aaf8cc853'),  
    '2': ObjectId('696d664a7431812aaf8cc854'),  
    '3': ObjectId('696d664a7431812aaf8cc855'),  
    '4': ObjectId('696d664a7431812aaf8cc856'),  
    '5': ObjectId('696d664a7431812aaf8cc857'),  
    '6': ObjectId('696d664a7431812aaf8cc858'),  
    '7': ObjectId('696d664a7431812aaf8cc859'),  
    '8': ObjectId('696d664a7431812aaf8cc85a'),  
    '9': ObjectId('696d664a7431812aaf8cc85b')  
  }  
}
```

```
> db.loginSessions.insertMany([
  {
    staffId: "S001",
    staffName: "Husin",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.10",
    loginTime: ISODate("2025-01-15T08:00:00Z")
  },
  {
    staffId: "S002",
    staffName: "Ika",
    role: "Nurse",
    status: "ACTIVE",
    ipAddress: "192.168.1.11",
    loginTime: ISODate("2025-01-15T08:10:00Z")
  },
  {
    staffId: "S003",
    staffName: "Alif",
    role: "Receptionist",
    status: "INACTIVE",
    ipAddress: "192.168.1.12",
    loginTime: ISODate("2025-01-01T09:00:00Z")
  },
```

```
{
  staffId: "S004",
  staffName: "Alia",
  role: "Pharmacist",
  status: "ACTIVE",
  ipAddress: "192.168.1.13",
  loginTime: ISODate("2025-01-15T08:20:00Z")
},
{
  staffId: "S005",
  staffName: "Nur",
  role: "Doctor",
  status: "INACTIVE",
  ipAddress: "192.168.1.14",
  loginTime: ISODate("2025-01-02T10:00:00Z")
},
{
  staffId: "S006",
  staffName: "Hafiz",
  role: "Receptionist",
  status: "ACTIVE",
  ipAddress: "192.168.1.15",
  loginTime: ISODate("2025-01-15T08:30:00Z")
},
```

```
{
  staffId: "S007",
  staffName: "Hadif",
  role: "Pharmacist",
  status: "ACTIVE",
  ipAddress: "192.168.1.16",
  loginTime: ISODate("2025-01-15T08:40:00Z")
},
{
  staffId: "S008",
  staffName: "Lee",
  role: "Doctor",
  status: "ACTIVE",
  ipAddress: "192.168.1.17",
  loginTime: ISODate("2025-01-03T11:00:00Z")
},
```

```
{
  staffId: "S009",
  staffName: "Farra",
  role: "Nurse",
  status: "INACTIVE",
  ipAddress: "192.168.1.18",
  loginTime: ISODate("2025-01-15T08:50:00Z")
},
{
  staffId: "S010",
  staffName: "Qawie",
  role: "Doctor",
  status: "ACTIVE",
  ipAddress: "192.168.1.19",
  loginTime: ISODate("2025-01-15T09:00:00Z")
}
])
```

```
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('696d690b7431812aaf8cc85c'),  
    '1': ObjectId('696d690b7431812aaf8cc85d'),  
    '2': ObjectId('696d690b7431812aaf8cc85e'),  
    '3': ObjectId('696d690b7431812aaf8cc85f'),  
    '4': ObjectId('696d690b7431812aaf8cc860'),  
    '5': ObjectId('696d690b7431812aaf8cc861'),  
    '6': ObjectId('696d690b7431812aaf8cc862'),  
    '7': ObjectId('696d690b7431812aaf8cc863'),  
    '8': ObjectId('696d690b7431812aaf8cc864'),  
    '9': ObjectId('696d690b7431812aaf8cc865')  
  }  
}
```

Part B

Question B1

```
> db.appointments.find({status: "COMPLETED", fee: {$gte: 100, $lte: 300}},  
  {_id: 0, appointmentId: 1, patientName: 1, department: 1, fee: 1}).sort({department: 1, fee: -1})
```

```
< {  
  appointmentId: 'A006',  
  patientName: 'Arisa',  
  department: 'Cardiology',  
  fee: 100  
}  
{  
  appointmentId: 'A003',  
  patientName: 'Nur',  
  department: 'Dermatology',  
  fee: 300  
}  
{  
  appointmentId: 'A008',  
  patientName: 'Nabil',  
  department: 'Neurology',  
  fee: 160  
}  
{  
  appointmentId: 'A005',  
  patientName: 'Farhan',  
  department: 'Neurology',  
  fee: 150  
}
```

```
{
  appointmentId: 'A001',
  patientName: 'Zara',
  department: 'Pediatrics',
  fee: 200
}
{
  appointmentId: 'A010',
  patientName: 'Tasha',
  department: 'Pediatrics',
  fee: 200
}
```

Question B2

```
> db.appointments.aggregate([{$group: {_id: "$department",  
totalAppointments: {$sum: 1}, totalFees: {$sum: "$fee"}}}, {$sort: {totalFees: -1}}])
```

```
< {  
  _id: 'Neurology',  
  totalAppointments: 4,  
  totalFees: 620  
}  
{  
  _id: 'Dermatology',  
  totalAppointments: 2,  
  totalFees: 600  
}  
{  
  _id: 'Pediatrics',  
  totalAppointments: 2,  
  totalFees: 400  
}  
{  
  _id: 'Cardiology',  
  totalAppointments: 2,  
  totalFees: 220  
}
```

Part C

Question C1

```
> db.loginSessions.find({status: "ACTIVE"},  
  {_id: 0, staffName: 1, role: 1, ipAddress: 1, loginTime: 1}).sort({loginTime: -1})
```

```
< {  
  staffName: 'Qawie',  
  role: 'Doctor',  
  ipAddress: '192.168.1.19',  
  loginTime: 2025-01-15T09:00:00.000Z  
}  
{  
  staffName: 'Hadif',  
  role: 'Pharmacist',  
  ipAddress: '192.168.1.16',  
  loginTime: 2025-01-15T08:40:00.000Z  
}  
{  
  staffName: 'Hafiz',  
  role: 'Receptionist',  
  ipAddress: '192.168.1.15',  
  loginTime: 2025-01-15T08:30:00.000Z  
}  
{  
  staffName: 'Alia',  
  role: 'Pharmacist',  
  ipAddress: '192.168.1.13',  
  loginTime: 2025-01-15T08:20:00.000Z  
}
```

```
{
  staffName: 'Ika',
  role: 'Nurse',
  ipAddress: '192.168.1.11',
  loginTime: 2025-01-15T08:10:00.000Z
}
{
  staffName: 'Husin',
  role: 'Doctor',
  ipAddress: '192.168.1.10',
  loginTime: 2025-01-15T08:00:00.000Z
}
{
  staffName: 'Lee',
  role: 'Doctor',
  ipAddress: '192.168.1.17',
  loginTime: 2025-01-03T11:00:00.000Z
}
```

Question C2

(a)

```
> db.loginSessions.distinct("ipAddress", {status: "ACTIVE"})
```

```
< [  
  '192.168.1.10',  
  '192.168.1.11',  
  '192.168.1.13',  
  '192.168.1.15',  
  '192.168.1.16',  
  '192.168.1.17',  
  '192.168.1.19'  
]
```

(b)

```
> db.loginSessions.aggregate([{$match: {status: "ACTIVE"}}, {$group: {_id: "$staffId"},  
  {$count: "totalActiveStaff"}}])
```

```
< {  
  totalActiveStaff: 7  
}
```

Part D

Question D1

```
> db.appointments.createIndex({department: 1, status: 1})
```

```
< department_1_status_1
```

```
> db.appointments.getIndexes()
```

```
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { department: 1, status: 1 },
    name: 'department_1_status_1'
  }
]
```

Question D2

(a)

```
> db.appointments.updateMany({status: "CANCELLED"}, {$set: {status: "ARCHIVED"}})
```

```
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 2,  
  modifiedCount: 2,  
  upsertedCount: 0  
}
```

(b)

```
> db.loginSessions.deleteMany({status: "INACTIVE"})
```

```
< {  
  acknowledged: true,  
  deletedCount: 3  
}
```